1-prop-logic: 命题逻辑 (Propositional Logic)

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“再回到语法”

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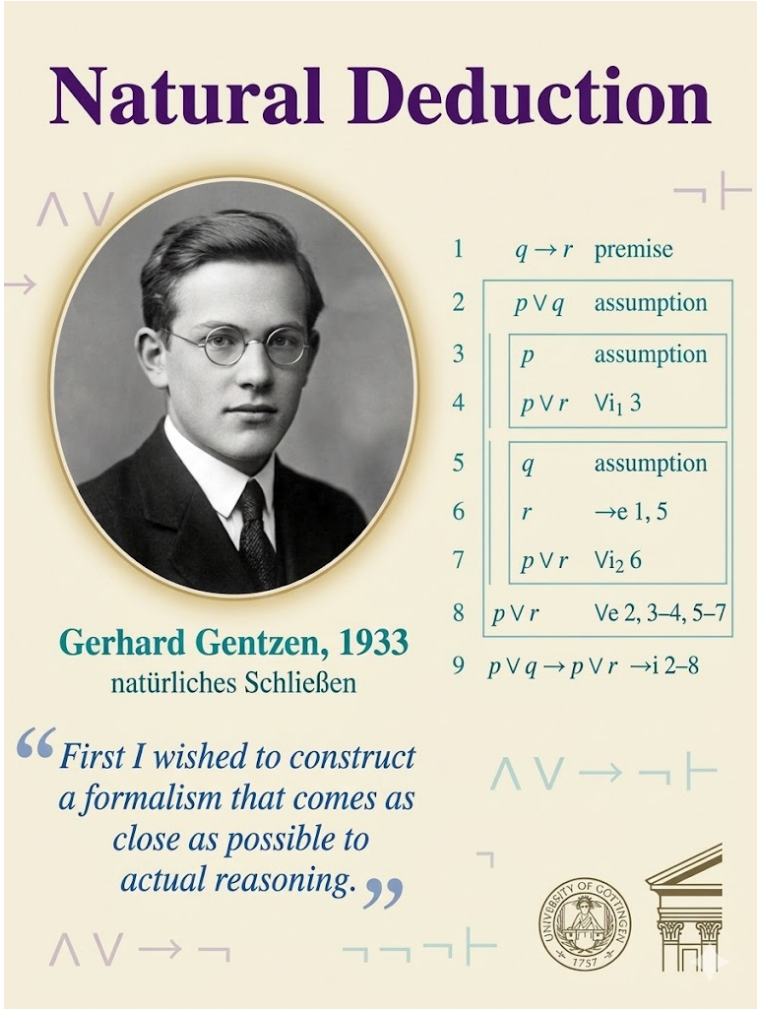
暂时**忘掉语义**!!! 一起来研究公式之间的**推导**关系

命题逻辑的自然推理 (演绎; 推演) 系统



$$\boxed{\dfrac{\alpha \qquad \beta \qquad \ldots \quad (\text{前提})}{\gamma \quad (\text{结论})} \qquad (\text{规则名称})}$$

$$\{\alpha, \beta, \ldots\} \vdash \gamma$$

从上往下看: 展示证明过程

$$\boxed{\dfrac{\alpha \qquad \beta \qquad \ldots \quad (\text{前提})}{\gamma \quad (\text{结论})} \quad (\text{规则名称})}$$

$$\{\alpha, \beta, \ldots\} \vdash \gamma$$

从上往下看: 展示证明过程

$$\boxed{\dfrac{\alpha \qquad \beta \qquad \ldots \quad (\text{前提})}{\gamma \quad (\text{结论})} \quad (\text{规则名称})}$$

$$\{\alpha, \beta, \ldots\} \vdash \gamma$$

从下往上看: 提供证明思路

	introduction	*elimination*
$\wedge$	$\dfrac{\phi \quad \psi}{\phi \wedge \psi}\,\wedge\mathrm{i}$	$\dfrac{\phi \wedge \psi}{\phi}\,\wedge\mathrm{e}_1 \qquad \dfrac{\phi \wedge \psi}{\psi}\,\wedge\mathrm{e}_2$
$\vee$	$\dfrac{\phi}{\phi \vee \psi}\,\vee\mathrm{i}_1 \qquad \dfrac{\psi}{\phi \vee \psi}\,\vee\mathrm{i}_2$	$\dfrac{\phi \vee \psi \quad \boxed{\begin{array}{c}\phi\\ \vdots\\ \chi\end{array}} \quad \boxed{\begin{array}{c}\psi\\ \vdots\\ \chi\end{array}}}{\chi}\,\vee\mathrm{e}$
$\rightarrow$	$\dfrac{\boxed{\begin{array}{c}\phi\\ \vdots\\ \psi\end{array}}}{\phi \boxed{\rightarrow} \psi}\,\rightarrow\mathrm{i}$	$\dfrac{\phi \quad \phi \boxed{\rightarrow} \psi}{\psi}\,\rightarrow\mathrm{e}$
$\neg$	$\dfrac{\boxed{\begin{array}{c}\phi\\ \vdots\\ \bot\end{array}}}{\neg\phi}\,\neg\mathrm{i}$	$\dfrac{\phi \quad \neg\phi}{\bot}\,\neg\mathrm{e}$
$\bot$	(no introduction rule for $\bot$)	$\dfrac{\bot}{\phi}\,\bot\mathrm{e}$
$\neg\neg$		$\dfrac{\neg\neg\phi}{\phi}\,\neg\neg\mathrm{e}$

$\wedge$

$$\frac{P \qquad Q}{P \wedge Q} \quad (\wedge\text{-intro})$$

$\wedge$

$$\frac{P \qquad Q}{P \wedge Q} \quad (\wedge\text{-intro})$$

$$\frac{P \wedge Q}{P} \quad (\wedge\text{-elim-left})$$

$\wedge$

$$\frac{P \qquad Q}{P \wedge Q} \quad (\wedge\text{-intro})$$

$$\frac{P \wedge Q}{P} \quad (\wedge\text{-elim-left})$$

$$\frac{P \wedge Q}{Q} \quad (\wedge\text{-elim-right})$$

“∧” 推理规则的应用

$$\{p \wedge q, r\} \vdash q \wedge r$$

Proof.

1.
2.
3.
4.

□

“∧” 推理规则的应用

$$\{p \wedge q, r\} \vdash q \wedge r$$

Proof.

1.	$p \wedge q$	premise
2.	r	premise
3.		
4.		

□

“∧” 推理规则的应用

$$\{p \wedge q, r\} \vdash q \wedge r$$

Proof.

1.	$p \wedge q$	premise
2.	r	premise
3.		
4.	$q \wedge r$	

□

“∧” 推理规则的应用

$$\{p \wedge q, r\} \vdash q \wedge r$$

Proof.

1.	$p \wedge q$	premise
2.	r	premise
3.	q	$\wedge$-elim$_r$ 1
4.	$q \wedge r$	

□

“∧” 推理规则的应用

$$\{p \wedge q, r\} \vdash q \wedge r$$

Proof.

1.	$p \wedge q$	premise
2.	r	premise
3.	q	$\wedge$-elim$_r$ 1
4.	$q \wedge r$	$\wedge$-intro 3,2

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.
2.
3.
4.
5.
6.

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.		
4.		
5.		
6.		

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.		
4.		
5.		
6.	$q \wedge s$	

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	$p \wedge q$	$\wedge$-elim$_l$ 1
4.		
5.		
6.	$q \wedge s$	

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	$p \wedge q$	$\wedge$-elim$_l$ 1
4.	q	$\wedge$-elim$_r$ 3
5.		
6.	$q \wedge s$	

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	$p \wedge q$	$\wedge$-elim$_l$ 1
4.	q	$\wedge$-elim$_r$ 3
5.	s	$\wedge$-elim$_l$ 2
6.	$q \wedge s$	

□

“∧” 推理规则的应用

$$\{(p \wedge q) \wedge r, s \wedge t\} \vdash q \wedge s$$

Proof.

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	$p \wedge q$	$\wedge$-elim$_l$ 1
4.	q	$\wedge$-elim$_r$ 3
5.	s	$\wedge$-elim$_l$ 2
6.	$q \wedge s$	$\wedge$-intro 4, 5

□

¬¬

$$\frac{\alpha}{\neg\neg\alpha} \quad (\neg\neg\text{-intro})$$

$$\frac{\neg\neg\alpha}{\alpha} \quad (\neg\neg\text{-elem})$$

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.
2.
3.
4.
5.
6.

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.		
4.		
5.		
6.		

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.		
4.		
5.		
6.	$\neg\neg p \wedge r$	

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.	$\neg\neg p$	$\neg\neg$-intro 1
4.		
5.		
6.	$\neg\neg p \wedge r$	

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.	$\neg\neg p$	¬¬-intro 1
4.	$q \wedge r$	¬¬-elim 2
5.		
6.	$\neg\neg p \wedge r$	

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.	$\neg\neg p$	$\neg\neg$-intro 1
4.	$q \wedge r$	$\neg\neg$-elim 2
5.	r	$\wedge$-elim$_r$ 4
6.	$\neg\neg p \wedge r$	

□

“¬¬” 推理规则的应用

$$\{p, \neg\neg(q \wedge r)\} \vdash \neg\neg p \wedge r$$

Proof.

1.	p	premise
2.	$\neg\neg(q \wedge r)$	premise
3.	$\neg\neg p$	$\neg\neg$-intro 1
4.	$q \wedge r$	$\neg\neg$-elim 2
5.	r	$\wedge$-elim$_r$ 4
6.	$\neg\neg p \wedge r$	$\wedge$-intro 3,5

□

$\rightarrow$ -elim

$$\frac{\alpha \rightarrow \beta \qquad \alpha}{\beta} \quad (\rightarrow \text{-elim (MP: } \textit{modus ponens; } \text{肯定前件}))$$

$\rightarrow$ -elim

$$\frac{\alpha \rightarrow \beta \qquad \alpha}{\beta} \quad (\rightarrow \text{-elim (MP: } \textit{modus ponens; } \text{肯定前件}))$$

$$\frac{\alpha \rightarrow \beta \qquad \neg\beta}{\neg\alpha} \quad (\text{MT: } \textit{modus tollens; } \text{否定后件})$$

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.
2.
3.
4.
5.
6.

$\square$

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \to q$	premise
3.	p	premise
4.		
5.		
6.		

□

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \to q$	premise
3.	p	premise
4.		
5.		
6.	r	

□

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \to q$	premise
3.	p	premise
4.	$q \to r$	$\to$ -elim 1, 3
5.		
6.	r	

□

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \to q$	premise
3.	p	premise
4.	$q \to r$	$\to$ -elim 1, 3
5.	q	$\to$ -elim 2, 3
6.	r	

□

“$\to$ -elim ” 推理规则的应用

$$\{p, p \to q, p \to (q \to r)\} \vdash r$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \to q$	premise
3.	p	premise
4.	$q \to r$	$\to$ -elim 1, 3
5.	q	$\to$ -elim 2, 3
6.	r	$\to$ -elim 4, 5

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow (q \rightarrow r), p, \neg r\} \vdash \neg q$$

Proof.

1.
2.
3.
4.
5.

$\square$

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow (q \rightarrow r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.		
5.		

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow (q \rightarrow r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.		
5.	$\neg q$	

□

“$\to$ -elim ” 推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	$q \to r$	$\to$ -elim 1, 2
5.	$\neg q$	

□

“$\to$ -elim ” 推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	$q \to r$	$\to$ -elim 1, 2
5.	$\neg q$	MT 4, 3

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{\neg p \rightarrow q, \neg q\} \vdash p$$

Proof.

1.
2.
3.
4.

$\square$

“$\rightarrow$ -elim ” 推理规则的应用

$$\{\neg p \rightarrow q, \neg q\} \vdash p$$

Proof.

1.	$\neg p \rightarrow q$	premise
2.	$\neg q$	premise
3.		
4.		

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{\neg p \rightarrow q, \neg q\} \vdash p$$

Proof.

1.	$\neg p \rightarrow q$	premise
2.	$\neg q$	premise
3.		
4.	p	

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{\neg p \rightarrow q, \neg q\} \vdash p$$

Proof.

1.	$\neg p \rightarrow q$	premise
2.	$\neg q$	premise
3.	$\neg\neg p$	MT 1, 2
4.	p	

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{\neg p \rightarrow q, \neg q\} \vdash p$$

Proof.

1.	$\neg p \rightarrow q$	premise
2.	$\neg q$	premise
3.	$\neg\neg p$	MT 1, 2
4.	p	$\neg\neg$-elim 3

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow \neg q, q\} \vdash \neg p$$

Proof.

1.
2.
3.
4.

$\square$

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow \neg q, q\} \vdash \neg p$$

Proof.

1.	$p \rightarrow \neg q$	premise
2.	q	premise
3.		
4.		

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow \neg q, q\} \vdash \neg p$$

Proof.

1.	$p \rightarrow \neg q$	premise
2.	q	premise
3.		
4.	$\neg p$	

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow \neg q, q\} \vdash \neg p$$

Proof.

1.	$p \rightarrow \neg q$	premise
2.	q	premise
3.	$\neg\neg q$	$\neg\neg$-intro 2
4.	$\neg p$	

□

“$\rightarrow$ -elim ” 推理规则的应用

$$\{p \rightarrow \neg q, q\} \vdash \neg p$$

Proof.

1.	$p \rightarrow \neg q$	premise
2.	q	premise
3.	$\neg\neg q$	$\neg\neg$-intro 2
4.	$\neg p$	MT 1, 3

□

$\rightarrow$ -intro

$$\frac{\begin{array}{c}\alpha \\ \vdots \\ \beta\end{array}}{\alpha \rightarrow \beta} \quad (\rightarrow \text{-intro})$$

$\rightarrow$ -intro

$$\frac{\begin{array}{c}\alpha \\ \vdots \\ \beta\end{array}}{\alpha \rightarrow \beta} \quad (\rightarrow \text{-intro})$$

The assumption α is (and should be) discharged.

“$\rightarrow$ -intro ” 推理规则的应用

$$\{\neg q \rightarrow \neg p\} \vdash p \rightarrow \neg\neg q$$

Proof.

1.

2.

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\{\neg q \rightarrow \neg p\} \vdash p \rightarrow \neg\neg q$$

Proof.

1.	$\neg q \rightarrow \neg p$	premise
2.		

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\{\neg q \rightarrow \neg p\} \vdash p \rightarrow \neg\neg q$$

Proof.

1.	$\neg q \rightarrow \neg p$	premise
2.		

$$p \rightarrow \neg\neg q$$

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{\neg q \rightarrow \neg p\} \vdash p \rightarrow \neg\neg q$$

Proof.

1.	$\neg q \rightarrow \neg p$	premise
2.	p	assumption
3.		
4.		
5.	$p \rightarrow \neg\neg q$	

□

“$\to$ -intro ” 推理规则的应用

$$\{\neg q \to \neg p\} \vdash p \to \neg\neg q$$

Proof.

1.	$\neg q \to \neg p$	premise
2.	p	assumption
3.		
4.	$\neg\neg q$	
5.	$p \to \neg\neg q$	

□

“$\to$ -intro ” 推理规则的应用

$$\{\neg q \to \neg p\} \vdash p \to \neg\neg q$$

Proof.

1.	$\neg q \to \neg p$	premise
2.	p	assumption
3.	$\neg\neg p$	$\neg\neg$-intro 2
4.	$\neg\neg q$	
5.	$p \to \neg\neg q$	

□

“$\to$ -intro ” 推理规则的应用

$$\{\neg q \to \neg p\} \vdash p \to \neg\neg q$$

Proof.

1.	$\neg q \to \neg p$	premise
2.	p	assumption
3.	$\neg\neg p$	$\neg\neg$-intro 2
4.	$\neg\neg q$	MT 1, 3
5.	$p \to \neg\neg q$	

□

“$\to$ -intro ” 推理规则的应用

$$\{\neg q \to \neg p\} \vdash p \to \neg\neg q$$

Proof.

1.	$\neg q \to \neg p$	premise
2.	p	assumption
3.	$\neg\neg p$	$\neg\neg$-intro 2
4.	$\neg\neg q$	MT 1, 3
5.	$p \to \neg\neg q$	$\to$ -intro 2–4

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.

$\square$

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.

$$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.		
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.		
	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.		
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.		
	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.		
5.		
6.		
7.		
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.		
5.		
6.		
7.	r	
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.		
6.		
7.	r	
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.		
7.	r	
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.	q	$\neg\neg$-elim 5
7.	r	
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.	q	$\neg\neg$-elim 5
7.	r	$\to$ -elim 1, 6
8.	$p \to r$	
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.	q	$\neg\neg$-elim 5
7.	r	$\to$ -elim 1, 6
8.	$p \to r$	$\to$ -intro 3–7
9.	$(\neg q \to \neg p) \to (p \to r)$	
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.	q	$\neg\neg$-elim 5
7.	r	$\to$ -elim 1, 6
8.	$p \to r$	$\to$ -intro 3–7
9.	$(\neg q \to \neg p) \to (p \to r)$	$\to$ -intro 2–8
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$$

Proof.

1.	$q \to r$	assumption
2.	$\neg q \to \neg p$	assumption
3.	p	assumption
4.	$\neg\neg p$	$\neg\neg$-intro 3
5.	$\neg\neg q$	MT 2, 4
6.	q	$\neg\neg$-elim 5
7.	r	$\to$ -elim 1, 6
8.	$p \to r$	$\to$ -intro 3–7
9.	$(\neg q \to \neg p) \to (p \to r)$	$\to$ -intro 2–8
10.	$(q \to r) \to ((\neg q \to \neg p) \to (p \to r))$	$\to$ -intro 1–9

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \rightarrow r\} \vdash p \rightarrow (q \rightarrow r)}$$

Proof.

1.

2.

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \rightarrow r\} \vdash p \rightarrow (q \rightarrow r)}$$

Proof.

1.	$(p \wedge q) \rightarrow r$	premise
2.		

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \rightarrow r\} \vdash p \rightarrow (q \rightarrow r)}$$

Proof.

1.	$(p \wedge q) \rightarrow r$	premise
2.		

$$p \rightarrow (q \rightarrow r)$$

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \wedge q) \to r$	premise
2.	p	assumption
3.		
7.	$p \to (q \to r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \rightarrow r\} \vdash p \rightarrow (q \rightarrow r)}$$

Proof.

1.	$(p \wedge q) \rightarrow r$	premise
2.	p	assumption
3.		
	$q \rightarrow r$	
7.	$p \rightarrow (q \rightarrow r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \wedge q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.		
5.		
6.	$q \to r$	
7.	$p \to (q \to r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \wedge q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.		
5.	r	
6.	$q \to r$	
7.	$p \to (q \to r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \wedge q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.	$p \wedge q$	$\wedge$-intro 2, 3
5.	r	
6.	$q \to r$	
7.	$p \to (q \to r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \wedge q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \wedge q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.	$p \wedge q$	$\wedge$-intro 2, 3
5.	r	$\to$ -elim 1, 4
6.	$q \to r$	
7.	$p \to (q \to r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \land q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \land q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.	$p \land q$	$\land$-intro 2, 3
5.	r	$\to$ -elim 1, 4
6.	$q \to r$	$\to$ -intro 3–5
7.	$p \to (q \to r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{(p \land q) \to r\} \vdash p \to (q \to r)}$$

Proof.

1.	$(p \land q) \to r$	premise
2.	p	assumption
3.	q	assumption
4.	$p \land q$	$\land$-intro 2, 3
5.	r	$\to$ -elim 1, 4
6.	$q \to r$	$\to$ -intro 3–5
7.	$p \to (q \to r)$	$\to$ -intro 2–6

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{p \rightarrow (q \rightarrow r)\} \vdash (p \wedge q) \rightarrow r}$$

Proof.

1.

2.

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{p \rightarrow (q \rightarrow r)\} \vdash (p \wedge q) \rightarrow r}$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.		

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{p \rightarrow (q \rightarrow r)\} \vdash (p \wedge q) \rightarrow r}$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.		

$$(p \wedge q) \rightarrow r$$

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{p \to (q \to r)\} \vdash (p \wedge q) \to r}$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \wedge q$	assumption
3.		
4.		
5.		
6.		
7.	$(p \wedge q) \to r$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{p \to (q \to r)\} \vdash (p \wedge q) \to r}$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \wedge q$	assumption
3.		
4.		
5.		
6.	r	
7.	$(p \wedge q) \to r$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{p \rightarrow (q \rightarrow r)\} \vdash (p \wedge q) \rightarrow r}$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.	$p \wedge q$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.		
5.		
6.	r	
7.	$(p \wedge q) \rightarrow r$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{p \to (q \to r)\} \vdash (p \wedge q) \to r}$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \wedge q$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	q	$\wedge$-elim$_r$ 2
5.		
6.	r	
7.	$(p \wedge q) \to r$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\boxed{\{p \rightarrow (q \rightarrow r)\} \vdash (p \wedge q) \rightarrow r}$$

Proof.

1.	$p \rightarrow (q \rightarrow r)$	premise
2.	$p \wedge q$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	q	$\wedge$-elim$_r$ 2
5.	$q \rightarrow r$	$\rightarrow$ -elim 1, 3
6.	r	
7.	$(p \wedge q) \rightarrow r$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{p \to (q \to r)\} \vdash (p \wedge q) \to r}$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \wedge q$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	q	$\wedge$-elim$_r$ 2
5.	$q \to r$	$\to$ -elim 1, 3
6.	r	$\to$ -elim 5, 4
7.	$(p \wedge q) \to r$	

□

“$\to$ -intro ” 推理规则的应用

$$\boxed{\{p \to (q \to r)\} \vdash (p \wedge q) \to r}$$

Proof.

1.	$p \to (q \to r)$	premise
2.	$p \wedge q$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	q	$\wedge$-elim$_r$ 2
5.	$q \to r$	$\to$ -elim 1, 3
6.	r	$\to$ -elim 5, 4
7.	$(p \wedge q) \to r$	$\to$ -intro 2–6

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.

2.

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.		

$\square$

“$\to$ -intro ” 推理规则的应用

$$\{p \to q\} \vdash (p \wedge r) \to (q \wedge r)$$

Proof.

1.	$p \to q$	premise
2.		

$$(p \wedge r) \to (q \wedge r)$$

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.		
4.		
5.		
6.		
7.	$(p \wedge r) \rightarrow (q \wedge r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.		
4.		
5.		
6.	$q \wedge r$	
7.	$(p \wedge r) \rightarrow (q \wedge r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.		
4.		
5.	q	
6.	$q \wedge r$	
7.	$(p \wedge r) \rightarrow (q \wedge r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\{p \to q\} \vdash (p \wedge r) \to (q \wedge r)$$

Proof.

1.	$p \to q$	premise
2.	$p \wedge r$	assumption
3.		
4.	r	$\wedge$-elim$_r$ 2
5.	q	
6.	$q \wedge r$	
7.	$(p \wedge r) \to (q \wedge r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	r	$\wedge$-elim$_r$ 2
5.	q	
6.	$q \wedge r$	
7.	$(p \wedge r) \rightarrow (q \wedge r)$	

□

“$\to$ -intro ” 推理规则的应用

$$\{p \to q\} \vdash (p \wedge r) \to (q \wedge r)$$

Proof.

1.	$p \to q$	premise
2.	$p \wedge r$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	r	$\wedge$-elim$_r$ 2
5.	q	$\to$ -elim 1, 3
6.	$q \wedge r$	
7.	$(p \wedge r) \to (q \wedge r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	r	$\wedge$-elim$_r$ 2
5.	q	$\rightarrow$ -elim 1, 3
6.	$q \wedge r$	$\wedge$-intro 5, 4
7.	$(p \wedge r) \rightarrow (q \wedge r)$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\{p \rightarrow q\} \vdash (p \wedge r) \rightarrow (q \wedge r)$$

Proof.

1.	$p \rightarrow q$	premise
2.	$p \wedge r$	assumption
3.	p	$\wedge$-elim$_l$ 2
4.	r	$\wedge$-elim$_r$ 2
5.	q	$\rightarrow$ -elim 1, 3
6.	$q \wedge r$	$\wedge$-intro 5, 4
7.	$(p \wedge r) \rightarrow (q \wedge r)$	$\rightarrow$ -intro 2–6

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\vdash p \rightarrow p$$

Proof.

1.

$\square$

“$\rightarrow$ -intro ” 推理规则的应用

$$\vdash p \rightarrow p$$

Proof.

1.

$$p \rightarrow p$$

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\vdash p \rightarrow p$$

Proof.

1.	p	assumption
2.		
3.	$p \rightarrow p$	

□

“$\rightarrow$ -intro ” 推理规则的应用

$$\vdash p \rightarrow p$$

Proof.

1.	p	assumption
2.	p	copy 1
3.	$p \rightarrow p$	

□

“$\to$ -intro ” 推理规则的应用

$$\vdash p \to p$$

Proof.

1.	p	assumption
2.	p	copy 1
3.	$p \to p$	$\to$ -intro 1–2

□

$\vee$

$$\frac{\alpha}{\alpha \vee \beta} \quad (\vee\text{-intro-left})$$

$$\frac{\beta}{\alpha \vee \beta} \quad (\vee\text{-intro-right})$$

$\vee$

$$\frac{\alpha}{\alpha \vee \beta} \quad (\vee\text{-intro-left})$$

$$\frac{\beta}{\alpha \vee \beta} \quad (\vee\text{-intro-right})$$

$$\frac{\alpha \vee \beta \qquad \alpha \to \gamma \qquad \beta \to \gamma}{\gamma} \quad (\vee\text{-elim; (}\textit{case analysis; }\text{分情况讨论))}$$

$\vee$

$$\frac{\alpha}{\alpha \vee \beta} \quad (\vee\text{-intro-left})$$

$$\frac{\beta}{\alpha \vee \beta} \quad (\vee\text{-intro-right})$$

$$\frac{\alpha \vee \beta \qquad \alpha \to \gamma \qquad \beta \to \gamma}{\gamma} \quad (\vee\text{-elim; (}\textit{case analysis; 分情况讨论}\text{))}$$

$$\frac{\alpha \vee \beta \qquad \boxed{\begin{array}{c}\alpha \\ \vdots \\ \gamma\end{array}} \qquad \boxed{\begin{array}{c}\beta \\ \vdots \\ \gamma\end{array}}}{\gamma} \quad (\vee\text{-elim })$$

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.

2.

$\square$

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.		

□

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.		

$$q \vee p$$

$\square$

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.	p	assumption
3.		
4.		

$q \vee p$

$\square$

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.	p	assumption
3.	$q \vee p$	$\vee$-intro$_r$ 2
4.		

$$q \vee p$$

$\square$

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.	p	assumption
3.	$q \vee p$	$\vee$-intro$_r$ 2
4.	q	assumption
5.		
6.	$q \vee p$	

□

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.	p	assumption
3.	$q \vee p$	$\vee$-intro$_r$ 2
4.	q	assumption
5.	$q \vee p$	$\vee$-intro$_l$ 4
6.	$q \vee p$	

□

“$\vee$-elim ” 推理规则的应用

$$p \vee q \vdash q \vee p$$

Proof.

1.	$p \vee q$	premise
2.	p	assumption
3.	$q \vee p$	$\vee$-intro$_r$ 2
4.	q	assumption
5.	$q \vee p$	$\vee$-intro$_l$ 4
6.	$q \vee p$	$\vee$-elim 1, 2–3, 4–5

□

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.
2.

$\square$

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.		

$\square$

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.		

$$(p \vee q) \to (p \vee r)$$

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.		
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.		
	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.		
5.		
	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	
5.		
	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.		
	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.		
7.		
8.	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.		
7.	$p \vee r$	
8.	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“∨-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	r	$\to$ -elim 1, 5
7.	$p \vee r$	
8.	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	r	$\to$ -elim 1, 5
7.	$p \vee r$	$\vee$-intro$_r$ 6
8.	$p \vee r$	
9.	$(p \vee q) \to (p \vee r)$	

□

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	r	$\to$ -elim 1, 5
7.	$p \vee r$	$\vee$-intro$_r$ 6
8.	$p \vee r$	$\vee$-elim 2, 3–4, 5–7
9.	$(p \vee q) \to (p \vee r)$	

□

“$\vee$-elim ” 推理规则的应用

$$q \to r \vdash (p \vee q) \to (p \vee r)$$

Proof.

1.	$q \to r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	r	$\to$ -elim 1, 5
7.	$p \vee r$	$\vee$-intro$_r$ 6
8.	$p \vee r$	$\vee$-elim 2, 3–4, 5–7
9.	$(p \vee q) \to (p \vee r)$	$\to$ -intro 2–8

□

“$\vee$-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.
2.

$\square$

“$\vee$-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.		

$\square$

“$\vee$-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.		

$$p \vee (q \vee r)$$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.		
9.	$p \vee (q \vee r)$	

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.		
	$p \vee (q \vee r)$	
9.		
	$p \vee (q \vee r)$	

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.		
5.		
	$p \vee (q \vee r)$	
9.		

$p \vee (q \vee r)$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.		
	$p \vee (q \vee r)$	
9.		

$$p \vee (q \vee r)$$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.		
7.		
8.	$p \vee (q \vee r)$	
9.		

$p \vee (q \vee r)$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.		
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	
9.		

$p \vee (q \vee r)$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	$q \vee r$	$\vee$-intro$_l$ 5
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	
9.		

$p \vee (q \vee r)$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	$q \vee r$	$\vee$-intro$_l$ 5
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	$\vee$-elim 2, 3–4, 5–7
9.		

$p \vee (q \vee r)$

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	$q \vee r$	$\vee$-intro$_l$ 5
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	$\vee$-elim 2, 3–4, 5–7
9.	r	assumption
10.		
11.		
12.	$p \vee (q \vee r)$	

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	$q \vee r$	$\vee$-intro$_l$ 5
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	$\vee$-elim 2, 3–4, 5–7
9.	r	assumption
10.		
11.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 10
12.	$p \vee (q \vee r)$	

□

“∨-elim ” 推理规则的应用

$$(p \vee q) \vee r \vdash p \vee (q \vee r)$$

Proof.

1.	$(p \vee q) \vee r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee (q \vee r)$	$\vee$-intro$_l$ 3
5.	q	assumption
6.	$q \vee r$	$\vee$-intro$_l$ 5
7.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 6
8.	$p \vee (q \vee r)$	$\vee$-elim 2, 3–4, 5–7
9.	r	assumption
10.	$q \vee r$	$\vee$-intro$_r$ 9
11.	$p \vee (q \vee r)$	$\vee$-intro$_r$ 10
12.	$p \vee (q \vee r)$	

□

“∨-elim ” 推理规则的应用

$$(p \lor q) \lor r \vdash p \lor (q \lor r)$$

Proof.

1.	$(p \lor q) \lor r$	premise
2.	$p \lor q$	assumption
3.	p	assumption
4.	$p \lor (q \lor r)$	$\lor$-intro$_l$ 3
5.	q	assumption
6.	$q \lor r$	$\lor$-intro$_l$ 5
7.	$p \lor (q \lor r)$	$\lor$-intro$_r$ 6
8.	$p \lor (q \lor r)$	$\lor$-elim 2, 3–4, 5–7
9.	r	assumption
10.	$q \lor r$	$\lor$-intro$_r$ 9
11.	$p \lor (q \lor r)$	$\lor$-intro$_r$ 10
12.	$p \lor (q \lor r)$	$\lor$-elim 1, 2–8, 9–11

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.
2.
3.
4.

$\square$

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.		
3.		
4.		

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.		
3.		
4.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.		
4.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.		
6.		
7.		

$$(p \wedge q) \vee (p \wedge r)$$

$\square$

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.		
6.	$(p \wedge q) \vee (p \wedge r)$	
7.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	
7.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.		

$$(p \wedge q) \vee (p \wedge r)$$

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.	r	assumption
8.		
9.		
10.	$(p \wedge q) \vee (p \wedge r)$	

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.	r	assumption
8.		
9.	$(p \wedge q) \vee (p \wedge r)$	
10.	$(p \wedge q) \vee (p \wedge r)$	

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.	r	assumption
8.	$p \wedge r$	$\wedge$-intro 2, 7
9.	$(p \wedge q) \vee (p \wedge r)$	
10.	$(p \wedge q) \vee (p \wedge r)$	

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.	r	assumption
8.	$p \wedge r$	$\wedge$-intro 2, 7
9.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_r$ 8
10.	$(p \wedge q) \vee (p \wedge r)$	

□

$\vee$-elim 推理规则的应用

$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

Proof.

1.	$p \wedge (q \vee r)$	premise
2.	p	$\wedge$-elim$_l$ 1
3.	$q \vee r$	$\wedge$-elim$_r$ 1
4.	q	assumption
5.	$p \wedge q$	$\wedge$-intro 2, 4
6.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_l$ 5
7.	r	assumption
8.	$p \wedge r$	$\wedge$-intro 2, 7
9.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-intro$_r$ 8
10.	$(p \wedge q) \vee (p \wedge r)$	$\vee$-elim 3, 4–6, 7–9

□

$\vee$-elim 推理规则的应用 (请自行练习)

$$(p \wedge q) \vee (p \wedge r) \vdash p \wedge (q \vee r)$$

$\bot$

Definition (Contradiction ($\bot$; Falsum, Falsehood))

$\bot$ is a special formula of the form $\alpha \wedge \neg\alpha$ or $\neg\alpha \wedge \alpha$, where α is any formula.

TODO: $\neg$-intro

$$\frac{\alpha \qquad \neg\alpha}{\bot} \quad (\bot\text{-intro})$$

$$\frac{\bot}{\alpha} \quad (\bot\text{-elim}; \text{EFQ, ex falso quodlibet (Principle of Explosion)})$$

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.
2.

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.		

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.		

$$p \to q$$

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.		
7.		

$p \to q$

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.		
	$p \to q$	
7.		
	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.		
5.		
6.	$p \to q$	
7.		

$p \to q$

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.		
6.	$p \to q$	
7.		

$p \to q$

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	
7.		

$p \to q$

□

“⊥”-推理规则的应用

$$\neg p \lor q \vdash p \to q$$

Proof.

1.	$\neg p \lor q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.		

$p \to q$

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.		
11.	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \lor q \vdash p \to q$$

Proof.

1.	$\neg p \lor q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.		
	$p \to q$	
11.	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \vee q \vdash p \to q$$

Proof.

1.	$\neg p \vee q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.	p	assumption
9.		
10.	$p \to q$	
11.	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \lor q \vdash p \to q$$

Proof.

1.	$\neg p \lor q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.	p	assumption
9.	q	copy 7
10.	$p \to q$	
11.	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \lor q \vdash p \to q$$

Proof.

1.	$\neg p \lor q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.	p	assumption
9.	q	copy 7
10.	$p \to q$	$\to$ -intro 8–9
11.	$p \to q$	

□

“⊥”-推理规则的应用

$$\neg p \lor q \vdash p \to q$$

Proof.

1.	$\neg p \lor q$	premise
2.	$\neg p$	premise
3.	p	assumption
4.	$\bot$	$\neg$-elim 3, 2
5.	q	$\bot$-elim 4
6.	$p \to q$	$\to$ -intro 3–5
7.	q	premise
8.	p	assumption
9.	q	copy 7
10.	$p \to q$	$\to$ -intro 8–9
11.	$p \to q$	$\lor$-elim 1, 2–6, 7–10

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.
2.
3.

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.		

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.		

$\neg p$

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.		
5.		
6.		
7.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.		
5.		
6.	$\perp$	
7.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.	q	$\to$ -elim 1, 3
5.		
6.	$\bot$	
7.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.	q	$\to$ -elim 1, 3
5.	$\neg q$	$\to$ -elim 2, 3
6.	$\bot$	
7.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.	q	$\to$ -elim 1, 3
5.	$\neg q$	$\to$ -elim 2, 3
6.	$\bot$	$\neg$-elim 4, 5
7.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to q, p \to \neg q\} \vdash \neg p$$

Proof.

1.	$p \to q$	premise
2.	$p \to \neg q$	premise
3.	p	assumption
4.	q	$\to$-elim 1, 3
5.	$\neg q$	$\to$-elim 2, 3
6.	$\bot$	$\neg$-elim 4, 5
7.	$\neg p$	$\neg$-intro 3–6

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.

2.

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.		

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.		

$$\neg p$$

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.	p	assumption
3.		
4.		
5.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.	p	assumption
3.		
4.	$\bot$	
5.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.	p	assumption
3.	$\neg p$	$\to$ -elim 1, 2
4.	$\bot$	
5.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.	p	assumption
3.	$\neg p$	$\to$ -elim 1, 2
4.	$\bot$	$\neg$-elim 2, 3
5.	$\neg p$	

□

“¬”-推理规则的应用

$$\{p \to \neg p\} \vdash \neg p$$

Proof.

1.	$p \to \neg p$	premise
2.	p	assumption
3.	$\neg p$	$\to$ -elim 1, 2
4.	$\bot$	$\neg$-elim 2, 3
5.	$\neg p$	$\neg$-intro 2–4

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.
2.
3.
4.

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.		

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.		

$\neg q$

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.		
6.		
7.		
8.	$\neg q$	

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.		
6.		
7.	$\perp$	
8.	$\neg q$	

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.	$q \to r$	$\to$ -elim 1, 2
6.		
7.	$\bot$	
8.	$\neg q$	

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.	$q \to r$	$\to$-elim 1, 2
6.	r	$\to$-elim 5, 4
7.	$\bot$	
8.	$\neg q$	

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.	$q \to r$	$\to$-elim 1, 2
6.	r	$\to$-elim 5, 4
7.	$\perp$	$\neg$-elim 6, 3
8.	$\neg q$	

□

“¬”-推理规则的应用

$$\{p \to (q \to r), p, \neg r\} \vdash \neg q$$

Proof.

1.	$p \to (q \to r)$	premise
2.	p	premise
3.	$\neg r$	premise
4.	q	assumption
5.	$q \to r$	$\to$-elim 1, 2
6.	r	$\to$-elim 5, 4
7.	$\bot$	$\neg$-elim 6, 3
8.	$\neg q$	$\neg$-intro 4–7

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.
2.
3.
4.

9.

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.		
9.		

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.		
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.		
	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.		
6.		
7.		
8.	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.		
6.		
7.	$\perp$	
8.	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.	$p \wedge \neg q$	$\wedge$-intro 3, 4
6.		
7.	$\perp$	
8.	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.	$p \wedge \neg q$	$\wedge$-intro 3, 4
6.	r	$\to$-elim 1, 5
7.	$\perp$	
8.	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.	$p \wedge \neg q$	$\wedge$-intro 3, 4
6.	r	$\to$-elim 1, 5
7.	$\perp$	$\neg$-elim 6, 2
8.	$\neg\neg q$	
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.	$p \wedge \neg q$	$\wedge$-intro 3, 4
6.	r	$\to$-elim 1, 5
7.	$\bot$	$\neg$-elim 6, 2
8.	$\neg\neg q$	$\neg$-intro 4–7
9.	q	

□

“¬”-推理规则的应用

$$\{(p \wedge \neg q) \to r, \neg r, p\} \vdash q$$

Proof.

1.	$(p \wedge \neg q) \to r$	premise
2.	$\neg r$	premise
3.	p	premise
4.	$\neg q$	assumption
5.	$p \wedge \neg q$	$\wedge$-intro 3, 4
6.	r	$\to$-elim 1, 5
7.	$\bot$	$\neg$-elim 6, 2
8.	$\neg\neg q$	$\neg$-intro 4–7
9.	q	$\neg\neg$-elim 8

□

	introduction	*elimination*
$\wedge$	$\dfrac{\phi \quad \psi}{\phi \wedge \psi}\,\wedge\mathrm{i}$	$\dfrac{\phi \wedge \psi}{\phi}\,\wedge\mathrm{e}_1 \qquad \dfrac{\phi \wedge \psi}{\psi}\,\wedge\mathrm{e}_2$
$\vee$	$\dfrac{\phi}{\phi \vee \psi}\,\vee\mathrm{i}_1 \qquad \dfrac{\psi}{\phi \vee \psi}\,\vee\mathrm{i}_2$	$\dfrac{\phi \vee \psi \quad \boxed{\begin{array}{c}\phi\\ \vdots\\ \chi\end{array}} \quad \boxed{\begin{array}{c}\psi\\ \vdots\\ \chi\end{array}}}{\chi}\,\vee\mathrm{e}$
$\rightarrow$	$\dfrac{\boxed{\begin{array}{c}\phi\\ \vdots\\ \psi\end{array}}}{\phi \rightarrow \psi}\,\rightarrow\mathrm{i}$	$\dfrac{\phi \quad \phi \rightarrow \psi}{\psi}\,\rightarrow\mathrm{e}$
$\neg$	$\dfrac{\boxed{\begin{array}{c}\phi\\ \vdots\\ \bot\end{array}}}{\neg\phi}\,\neg\mathrm{i}$	$\dfrac{\phi \quad \neg\phi}{\bot}\,\neg\mathrm{e}$
$\bot$	(no introduction rule for $\bot$)	$\dfrac{\bot}{\phi}\,\bot\mathrm{e}$
$\neg\neg$		$\dfrac{\neg\neg\phi}{\phi}\,\neg\neg\mathrm{e}$

Some useful derived rules:

$$\frac{\phi \to \psi \quad \neg\psi}{\neg\phi}\ \text{MT} \qquad\qquad \frac{\phi}{\neg\neg\phi}\ \neg\neg\text{i}$$

$$\frac{\boxed{\begin{array}{c}\neg\phi\\ \vdots\\ \bot\end{array}}}{\phi}\ \text{PBC} \qquad\qquad \frac{}{\phi \vee \neg\phi}\ \text{LEM}$$

连这些推理规则也一并忘却吧!!!

使用命题逻辑进行推理

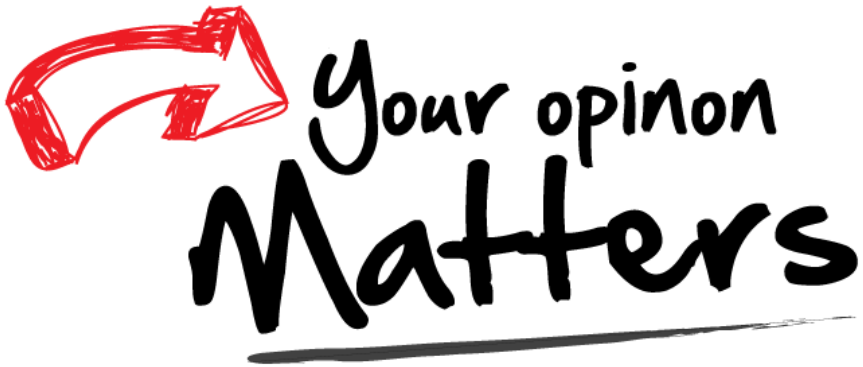
某公司要从赵、钱、孙、李、吴 5 名员工中选派某些人出国考察。由于某些不可描述的原因, 选派要求如下:

(1) 若赵去, 钱也去;
(2) 李、吴两人中必有一人去;
(3) 钱、孙两人中去且仅去一人;
(4) 孙、李两人同去或同不去;
(5) 若吴去, 则赵、钱也去;
(6) 只有孙去, 赵才会去。

(1) $Z \to Q$;
(2) $L \vee W$;
(3) $(Q \wedge \neg S) \vee (S \wedge \neg Q)$;
(4) $(S \wedge L) \vee (\neg S \wedge \neg L)$;
(5) $W \to Z \wedge Q$;
(6) $Z \to S$。

请使用形式化推理的方法帮该公司判断应选哪些人出国考察。

Thank You!



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